SECR 1213

Network Communication

Task 6B – Individual Report

Section: 01

Group: TTL Network Solutions

Lecturer: Dr. Mohd Kufaisal bin Mohd Sidik

Group members:

Name	Matric Number
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1.0 My Contribution

My primary responsibility was the Architectural and Logical Engineering of the First Floor, along with the Requirements Research and Core Hardware Selection.

- **Task 1**: I initiated the group name "TTL Network Solutions" and served as the lead architect for the First Floor Layout. I critically identified the need for infrastructure, adding a dedicated IDF Room and Electrical Room to the floor plan.
- **Task 2**: I formulated 6 key technical questions and answers regarding the network scope, user capacity, and application requirements. These answers provided the data foundation for our Feasibility analysis.
- **Task 3**: I led the research for the Core Network, recommending the Cisco ISR 1100 (SD-WAN) and Cisco Catalyst 9200L (740W PoE). I justified these enterprise-grade choices over consumer brands to ensure long-term stability.
- **Task 4**: I proposed using Fiber Optic cables for the vertical backbone. I designed the First Floor Network Distribution Diagrams and conducted the final review to ensure all diagrams matched the scaled floor plans.
- **Task 5**: I executed the VLSM IP Addressing for the entire First Floor, calculating specific subnets to prevent overlap with the Ground Floor.

2.0 Other Members' Contributions

- **Tey Xin Yi**: Ground Floor Lead. Designed the physical layout for the Ground Floor (Task 1) and wrote the Feasibility Study (Task 2). Researched WAPs (Task 3), designed Ground Floor diagrams (Task 4), and calculated Ground Floor IP subnets (Task 5).
- **Tan Hui Shan**: Documentation Lead. Prepared the Task 1 proposal and researched Firewalls (Task 3). Identified cabling types (Task 4), obtained the network address, and compiled the final Group Report.

3.0 Working as a Group

TTL Network Solutions operated on a Task-Distribution Model. To maximize efficiency, we split the building assignments: Tey handled the Ground Floor, and I handled the First Floor. This allowed us to work in parallel on AutoCAD designs without blocking each other. We utilized

Cross-Verification, where I checked Tey's calculations and she checked mine; this process helped us catch a scaling error in Task 4 before submission.

4.0 Learning Outcomes

From the Group: I learned the value of Standardization. Early on, we wasted time because we used different labels for our diagrams. We solved this by holding coordination meetings to agree on conventions (e.g., standardizing "IDF" vs "Server Room") before drafting.

From the Project:

- **Physical Constraints:** Task 1 and 4 taught me networking is physical. Switches require dedicated IDF rooms with specific power and cooling, not just logical connections.
- **Total Cost of Ownership:** Task 3 taught me that a cheap switch is useless if it lacks the PoE Budget for IP phones. Evaluating hardware on reliability (TCO) rather than sticker price is a critical skill.

5.0 Comments and Suggestions

The project was highly realistic. Dealing with a fixed budget (RM 2M) and a specific network address (172.16.4.0/23) forced us to make difficult engineering trade-offs.

I suggest utilizing AutoCAD templates provided by the faculty if possible. We spent significant time comparing tools before settling on AutoCAD; a standard template would save setup time.

6.0 Reflection on Every Task

- **Task 1**: Learned Architectural Planning; specifically, that identifying utility rooms (Electrical/Storage) is the first step in network design.
- **Task 2**: Learned Requirement Analysis; formulating research questions taught me that client needs dictate technical solutions.
- **Task 3**: Learned Technical Evaluation; how to read datasheets to find critical specs like "Fanless operation" or "FRU support."
- **Task 4**: Learned Scaling & Backbone Design; converting 2D drawings into real-world cable lengths and understanding why Fiber is mandatory for vertical runs.
- **Task 5**: Learned Network Segmentation; mastering VLSM to slice a single network block into efficient, secure subnets for different departments.